

REVENUE ASSURANCE — PROJECT MEMO

TO: KPC Revenue Assurance Stakeholders & PLP Supervisors

FROM: Null_Terminators Team

SUBJECT: Update — From Inuka Foundation Governance (Hackathon 2) to the Autonomous Supply Chain & Executive Control Plane (Hackathon 3)

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1. RECAP — HACKATHON 2: INUKA FOUNDATION GOVERNANCE

Hackathon 1 proved the Reconova data fabric on KPC's inbound fuel revenue: a three-way reconciliation of Dispatches → Invoices → Payments. Hackathon 2 re-pointed the same ETL pipeline, reconciliation logic, fraud graph, audit trail, and AI retrieval layer outward to govern Inuka Foundation beneficiary stipends (Attendance → Stipend Authorisation → Disbursement), with a consent ledger and full Kenya Data Protection Act alignment — proving the architecture is domain-agnostic, not a one-off build.

2. HACKATHON 3 — THE AUTONOMOUS SUPPLY CHAIN & EXECUTIVE CONTROL PLANE

For Hackathon 3 the platform was rebranded from FlowGuard to Reconova and pushed from passive reconciliation into an autonomous control plane. We selected Domain 4 — Supply Chain Revenue & Contract Compliance — solving Problem 8 (Loading Manifest vs. Invoiced Volume Reconciliation) as the primary problem statement, with Problem 7 (Automated Demurrage & Detention Billing Engine) integrated as an add-on.

- **Real-time gantry metering:** physical loading arms across Gantries 1–6 stream meter pulses that are reconciled against SAP billing invoices and bank receipts in real time to eliminate ghost loads and volume drift.
- **Autonomous gate lockout:** if physical volume exceeds invoiced volume beyond the statutory evaporation tolerance (PMS 0.30%, AGO 0.15%, IK/DPK 0.20%), the exit boom is held automatically pending an amended SAP billing note.
- **KRA iCMS tax adjustment gateway:** approved adjustment notes are queued and issued to KRA's e-Billing gateway automatically, with dead-letter-queue retry on failure.
- **3D isometric gantry bay matrix:** an interactive live HUD visualising all six loading bays, telemetry, and gate-barrier states.
- **Manager emergency override:** a secure, authenticated override protocol for emergency gate release, logged instantly to the immutable audit trail.

3. THE LINKUP — ONE DATA FABRIC, THREE STAGES

A single flow_direction discriminator and one shared reconciliation core now serve three missions on one codebase, one schema, and one set of quality gates:

Layer	Hackathon 1 — KPC	Hackathon 2 — Inuka	Hackathon 3 — Control Plane
Reconciliation	Dispatches → Invoices → Payments	Attendance → Stipend Auth → Disbursement	Real-time gantry metering vs. invoices
Control action	Post-facto anomaly detection	Fraud flag & notification	Automated physical gate lockout solenoid
Tax compliance	Manual KRA iCMS sync	N/A	Automated iCMS tax adjustment queueing
Visualization	Static KPI cards	Anomaly tables	3D isometric gantry bay matrix + HUD
Alerting	In-app notification inbox	Email alerts	PagerDuty / WhatsApp incident dispatch
Governance override	N/A	N/A	Authenticated manager gate override
AI assistant	KPC FAQ search	Multilingual RAG (EN/SW)	Unconditionally embedded RAG bot

4. QUANTIFIED IMPACT

Metric	KPC (Inbound, Stage 1)	Control Plane (Stage 3)
Reconciliation rate	95%+	100% real-time gantry gate auditing
Leakage / fraud detection	100% of volume-drift anomalies	0 unauthorized truck exits
Manual effort saved	80% (2 weeks/month → 2 days)	100% automated gate lockout action
Estimated savings	~200M KES/year (4B KES throughput, 5% baseline leakage)	Immediate tax recoupment via iCMS
Response latency	< 200ms	Sub-50ms lockout telemetry check

Honesty of assumptions: savings are modelled estimates — 5% baseline leakage on 4B KES KPC throughput — not guarantees.

5. QA, DEPLOYMENT & WHY THIS MATTERS FOR JUDGING

- Frontend: 8 test suites, 24 unit tests passing (100%). Backend: Pytest suite covering ETL, reconciliation, and control endpoints; CI/CD on every push via GitHub Actions.
- Bi-annual disaster recovery drill (14 Sept 2026): achieved RTO of 135 minutes (target < 240) and RPO of 18 minutes via WAL replay (target < 60); 63,240 audit records verified with zero broken pointers; Base Sepolia Merkle root match confirmed — all criteria PASS.
- Dual-layer cryptographic integrity: local SHA-256 hash-chain audit scan plus Base Sepolia blockchain Merkle-proof reconciliation, giving mathematical proof of zero post-disaster tampering.
- Live: revenue-assurance.fly.dev (API / Swagger) and reconovaa.vercel.app (frontend).

The Linkup is direct evidence for "Technical Quality & Integration": the same ETL, reconciliation, fraud-detection, alerting, audit, and AI-retrieval framework now also drives autonomous physical control — a gate lockout solenoid and automated tax adjustment — with no duplicated stack, turning a single hackathon build into a governed, self-enforcing platform spanning KPC revenue, Inuka Foundation disbursements, and now physical supply-chain compliance.

CONCLUSION

Hackathon 1 proved the model on inbound KPC revenue. Hackathon 2 proved it generalises to outbound Inuka Foundation stipends. Hackathon 3 proves it can act autonomously: the same data fabric now closes the loop from detection to physical enforcement at the gantry gate, in real time, under full cryptographic audit. One architecture, three domains, one defensible source of truth — from passive reconciliation to an autonomous executive control plane.